2024 MIT Science Bowl

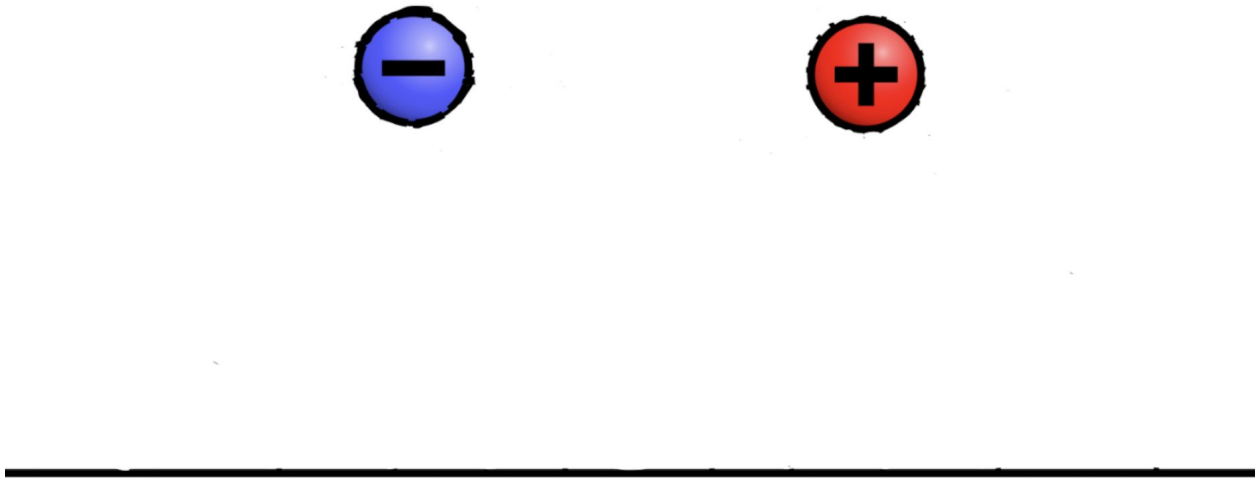
Round 14



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Bonus 1 (Physics)

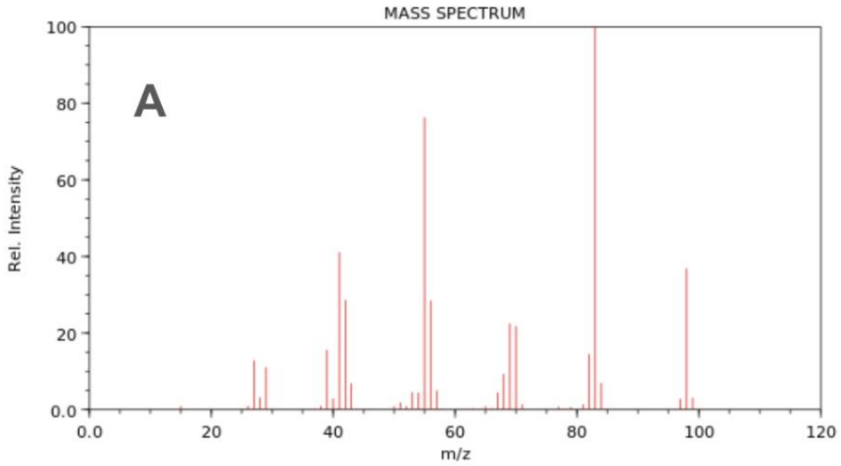


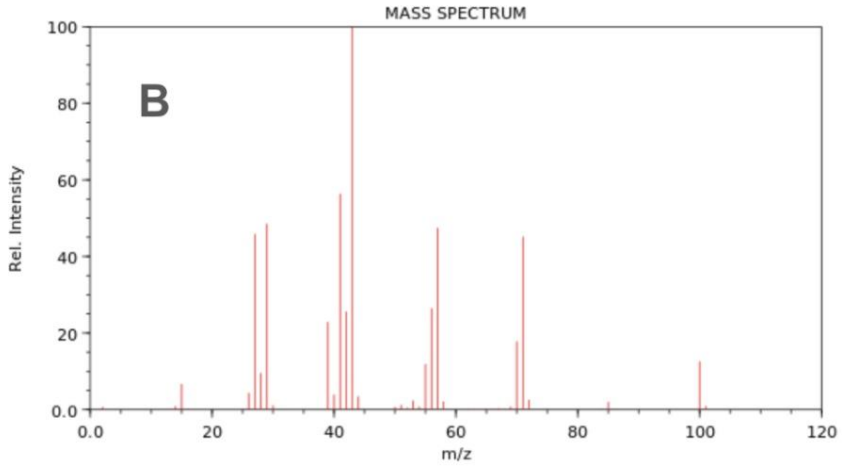


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Bonus 3 (Chem)


MASS SPECTRUM
A
Rel. Intensity
m/z
0.0
20
40
60
80
100
120

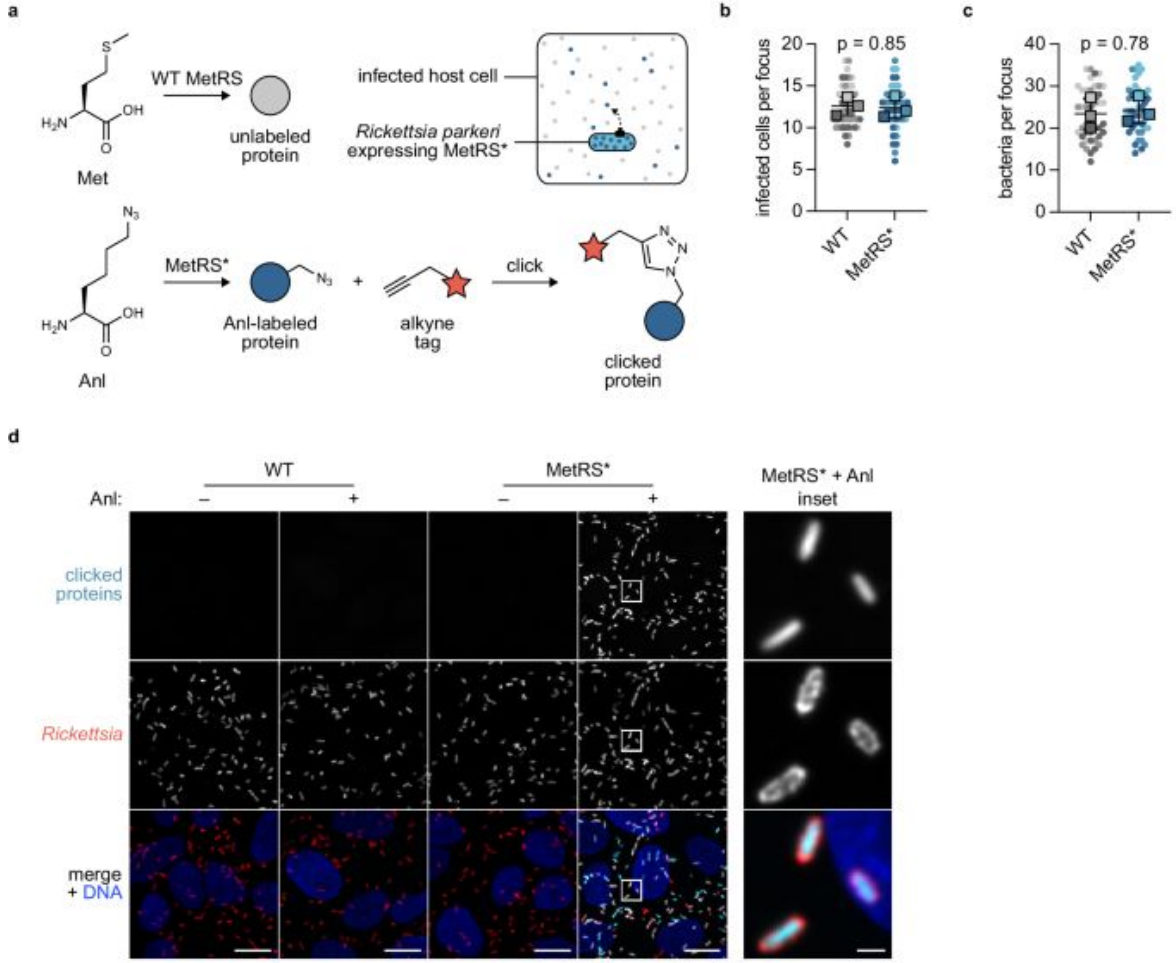

MASS SPECTRUM
B
Rel. Intensity
m/z
0.0
20
40
60
80
100
120



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Bonus 4 (Energy)

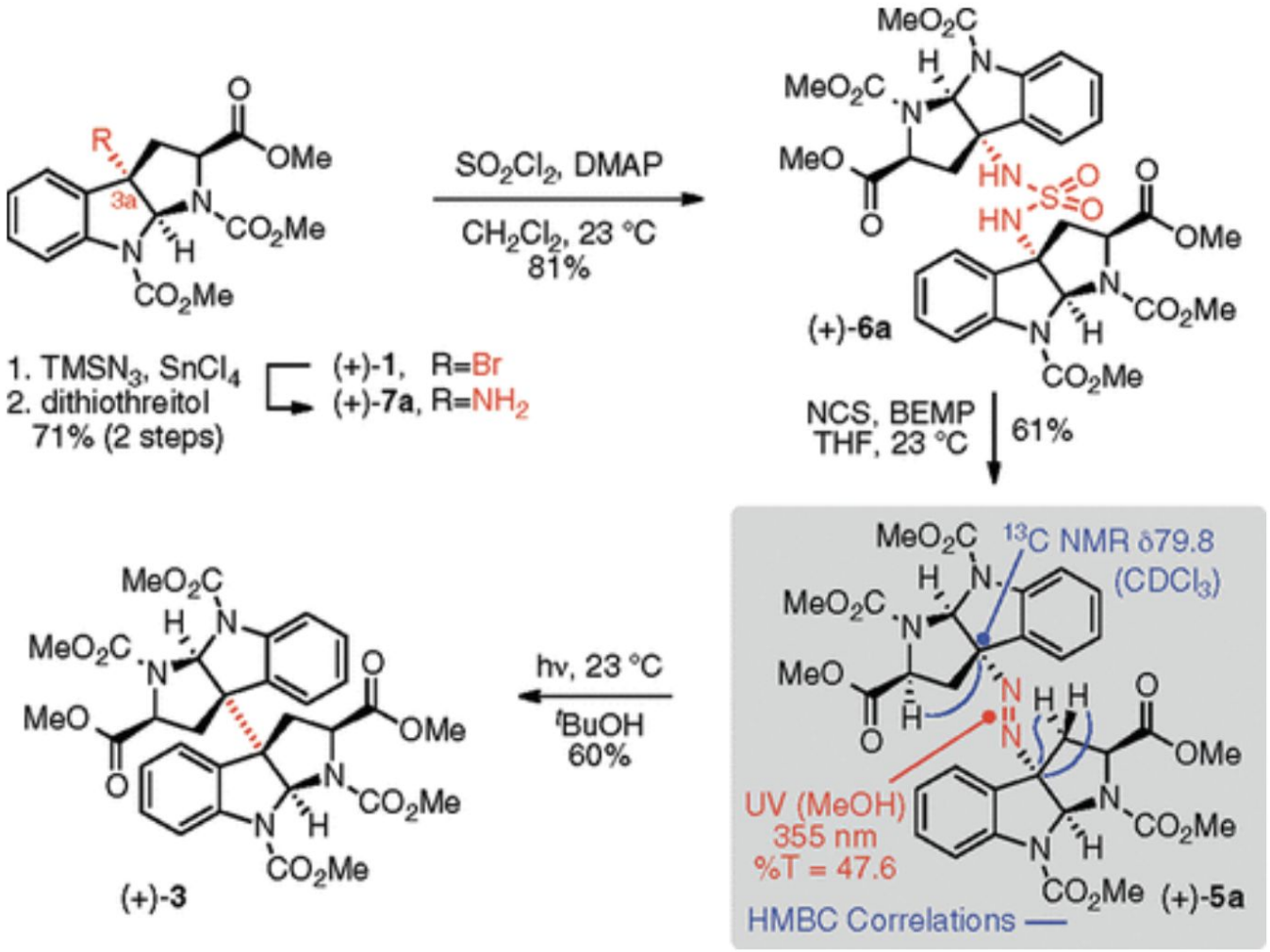

a
WT MetRS
unlabeled protein
Met
infected host cell
Rickettsia parkeri expressing MetRS*
MetRS*
Anl-labeled protein
alkyne tag
click
clicked protein
Anl
b
infected cells per focus
p = 0.85
WT
MetRS*
c
bacteria per focus
p = 0.78
WT
MetRS*
d
WT
MetRS*
MetRS* + Anl inset
Anl:
clicked proteins
Rickettsia
merge + DNA



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Bonus 7 (Energy)

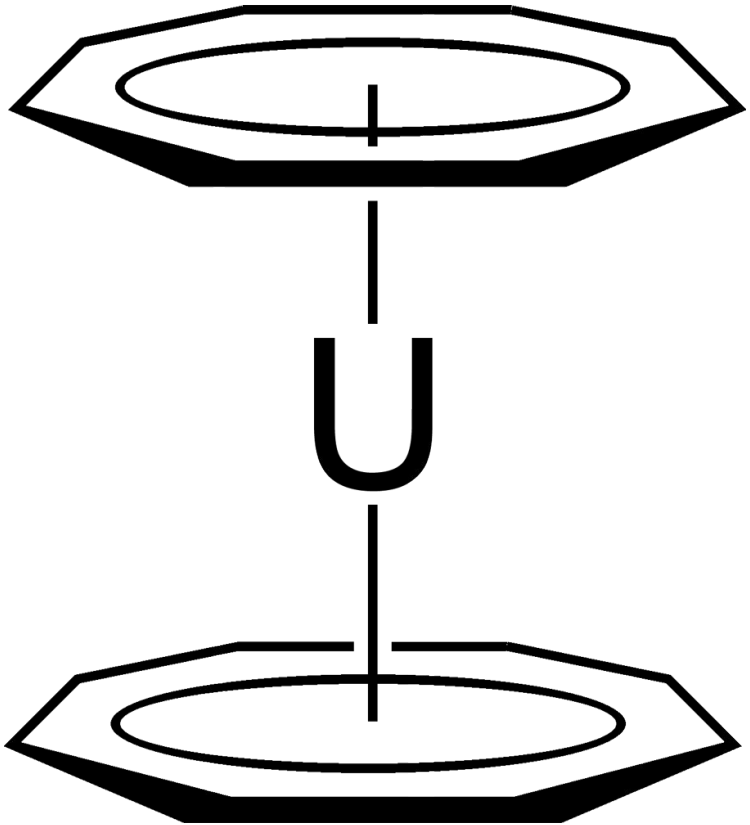

R
3a
OMe
CO_2Me
SO_2Cl_2, DMAP
CH_2Cl_2, 23 °C
81%
1. $TMSN_3$, $SnCl_4$
2. dithiothreitol
71% (2 steps)
(+)-1, R=Br
(+)-7a, R=NH_2
(+)-6a
NCS, BEMP
THF, 23 °C
61%
^{13}C NMR δ79.8
($CDCl_3$)
UV (MeOH)
355 nm
%T = 47.6
HMBC Correlations
(+)-5a
hv, 23 °C
^tBuOH
60%
(+)-3



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Bonus 8 (Chem)

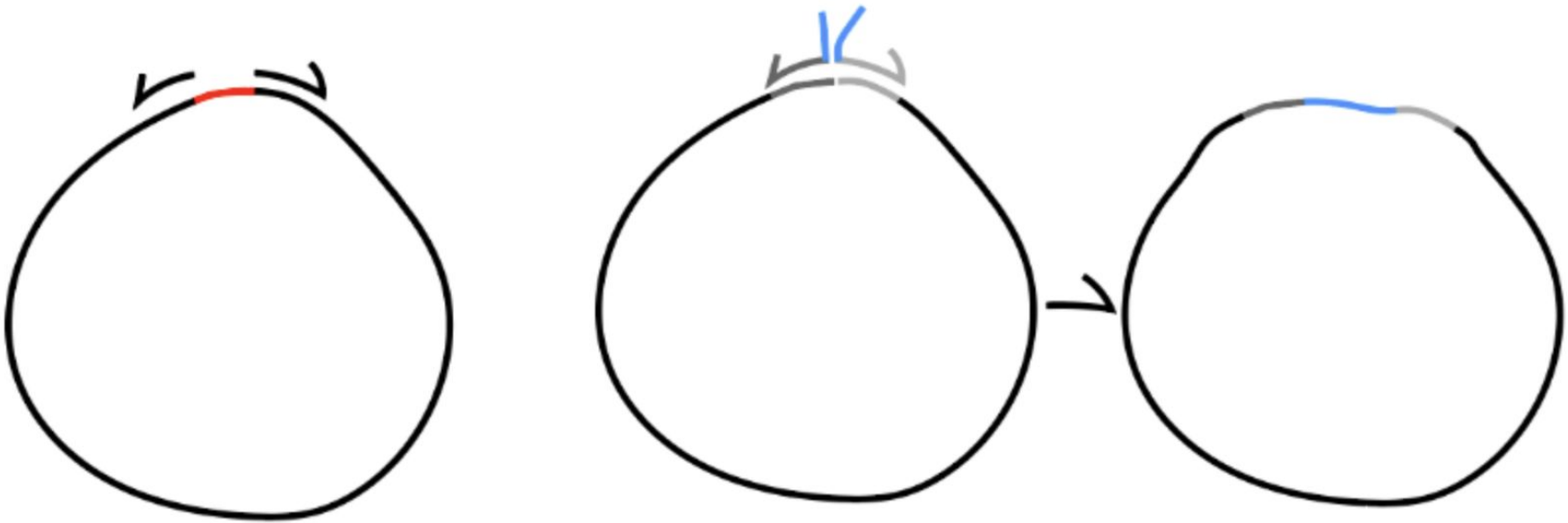

U



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Bonus 11 (Bio)

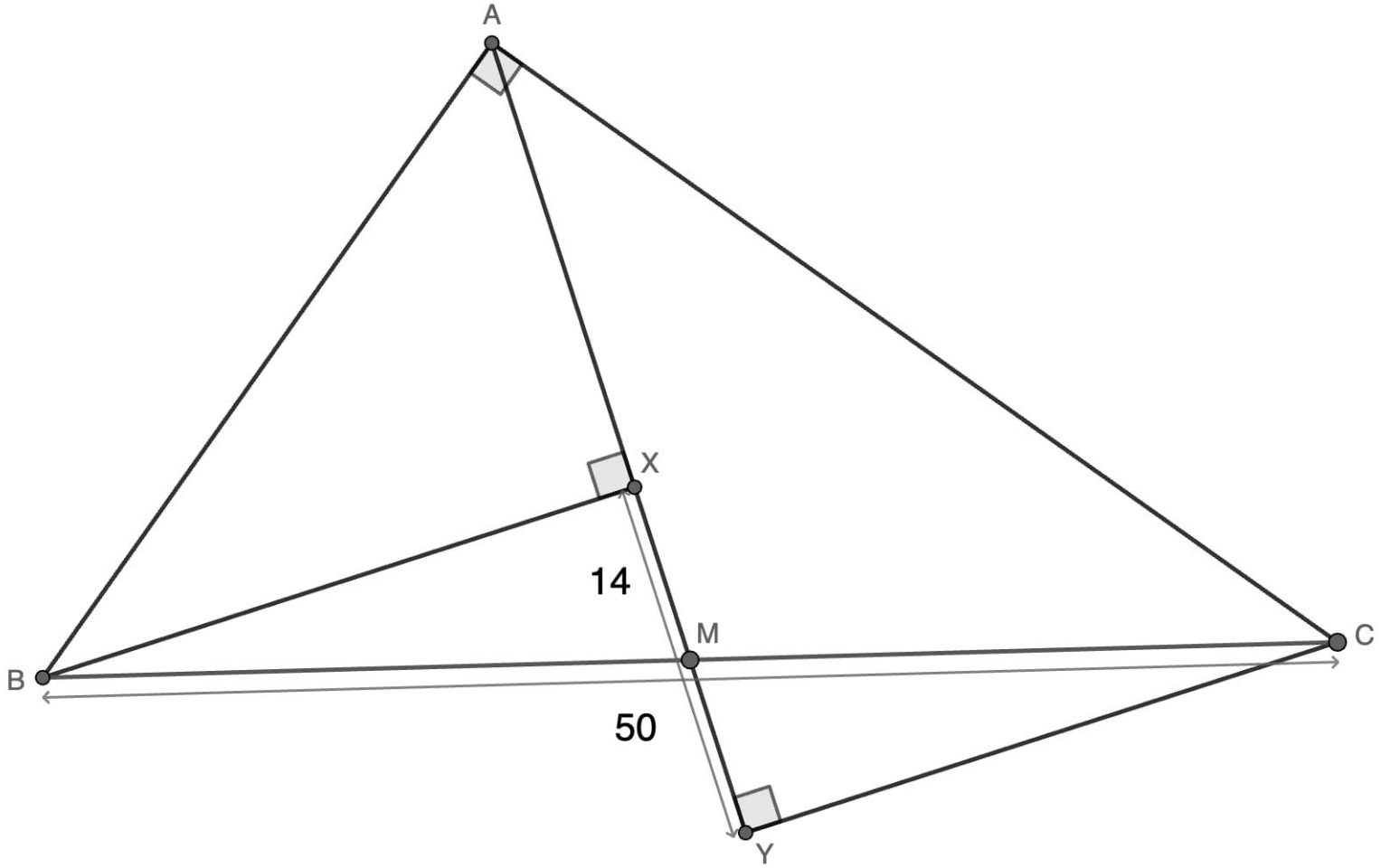




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Bonus 19 (Math)


A
X
14
M
C
B
50
Y



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